

Theory Problems (10 points each): Answer the following questions in 2-3 sentences each.

- 1. What is the primary disadvantage of using a brute-force approach like an exhaustive search for solving problems, particularly as the input size grows?**

The disadvantage is the high time complexity causing slow performance. That's because it processes all solutions so the larger the input, the more drastic the difference in speed is.

- 2. In the context of greedy algorithms, what does it mean to make a "locally optimal choice"?**

The locally optimal choice is picking the best method at the current stage. Something to note is that it doesn't take into account any future data, just what it sees now.

- 3. Explain why finding a counterexample is sufficient to prove that an algorithm is incorrect.**

Having an algorithm that is correct specifically needs to be accurate for every valid input. Therefore, if there is a valid input that outputs incorrectly, the algorithm is not correct.

- 4. Why is understanding the growth of functions important when analyzing the time complexity of algorithms?**

Knowing the growth of the algorithm is important for time complexity. The reason for this is the amount of time that it takes to complete the algorithm on the data set varies with its size. So in some cases, one function will be faster with smaller data sets while another would be faster with larger data sets and vice versa.

- 5. Suppose that for inputs of size n on a particular computer, insertion sort runs in $8n^2$ steps, and merge sort runs in $64n \lg n$ steps. For which values of n does insertion sort beat merge sort?**

insert merge

$$8n^2 \leq 64n \lg n$$

$$n^2 \leq 8n \lg n$$

$$n \leq 8 \lg n$$

n	$8 \lg n$	
1	0	X
2	8	✓
8	24	✓
16	32	✓
32	40	✓
64	48	X
50	45.15	X
40	42.58	X
45	43.94	X
44	43.67	X
43	43.41	✓

$$\therefore 2 \leq n \leq 43$$